CSA1735 – Artificial Intelligence

Assignment – CO1

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Design, Analyze and Validate an Efficient Search Strategy for a Warehouse Picking Robot

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*Code of Conduct: I**, Kaushik Narayanan V (192321047)**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

Scenario Recap

Autonomous robots in a fulfilment centre move between aisle junctions, cross-aisles, lift points and packing-station gates on a graph, with every move costing travel time. As new storage halls and mezzanines are added, the state space grows **1,000 → 10,000 → 100,000 → 1,000,000+** states, and the robot must still find a route from its charging dock to a requested bay. Any preprocessing (adjacency-list construction, duplicate detection) is included in the efficiency analysis. The Day-4 challenge adds a floor-marker reader fault, so the robot no longer knows exactly which of *k* junctions it is at.

TASK 1 – Design the Agent and the Search Algorithms (5 Marks)

PEAS description

Element	Specification
Performance measure	Minimum total travel-time cost to reach the requested bay/dock; safety (no collisions with other robots/staff); successful delivery within the order's time window
Environment	The warehouse floor graph: aisle junctions, cross-aisles, lift points, packing-station gates, other mobile robots and restocking staff
Actuators	Drive motors/wheels that move the robot along a chosen edge to an adjacent location
Sensors	Floor-marker (QR-code) reader for localisation, proximity/obstacle sensors, and a map of the known graph structure

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Environment classification, each justified against the scenario

Property	Classification	Justification
Observable	**Partially observable** (fully observable only when the QR reader works)	The robot perceives its own location via the floor marker and senses only nearby aisle occupancy, not the whole warehouse's current robot/staff positions at once
Deterministic	**Stochastic** (mostly deterministic moves, but outcomes affected by others)	A move to an adjacent junction normally succeeds, but an aisle may be blocked by another robot or restocking staff, changing the actual outcome
Episodic / Sequential	**Sequential**	Each move affects the robot's position and therefore the cost and options for every later move — routing is a connected sequence of decisions, not independent episodes
Static / Dynamic	**Dynamic**	Other robots and staff move and occupy aisles while the picking robot is still deliberating/travelling, so the environment changes independently of the robot's own actions
Discrete / Continuous	**Discrete**	Locations, actions (move to an adjacent junction) and time steps are all modelled as a finite graph with a countable action set, not continuous coordinates

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Property	Classification	Justification
Single / Multi-agent	**Multi-agent**	Other picking robots and human staff share the same floor and compete for the same aisles, so their presence must be accounted for (at least as dynamic obstacles)

View of AI the robot follows

The robot follows the **"acting rationally"** view: it is engineered to select the action sequence that maximises its performance measure (minimum travel-time cost, successful delivery) given what it currently perceives, rather than to mimic human reasoning ("acting/thinking humanly") or to formally model correct inference for its own sake ("thinking rationally"). It is a **rational problem-solving agent** because, for the (currently) fully-observable, deterministic, known, static, discrete, single-agent idealisation that search assumes, it computes the sequence of actions guaranteed to reach the goal at the lowest path cost before acting — the textbook definition of a rational agent for this environment class. **Relation to early AI history:** this is exactly the class of problem (route-finding over a discrete state graph) that motivated the earliest AI search work — from the General Problem Solver's means-ends analysis through to the uninformed graph-search algorithms (BFS, DFS, IDS) that remain the foundation before any domain-specific heuristic is added.

Problem formulation

Element	Definition
States	Each state is the robot's current location (an aisle junction, cross-aisle, lift point or gate) — a node in the warehouse graph
Initial state	The robot's charging dock (a specific node)
Actions	Move(location → adjacent_location) for every edge leaving the current node in the graph

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Element	Definition
Transition model	Result(s, Move(s→s')) = s' — moving along an edge deterministically changes the robot's location to the connected node (in the idealised, fully-observable search model; blocked aisles are handled as a dynamic re-planning trigger, outside the base search)
Goal test	s = requested_bay (or the dock, for a return trip)
Path cost	The sum of the travel-time costs of the edges traversed, g(n) = Σ cost(edge) along the path from the dock to the current node

Three meaningfully different uninformed search strategies

I selected **Breadth-First Search (BFS, iterative/queue-based)**, **Depth-First Search (DFS, iterative/stack-based)** and **Iterative Deepening Search (IDS, whose depth-limited core is recursive)**. These are not an iterative/recursive pair of the *same* algorithm: BFS expands shallowest-first with no backtracking of the frontier order, DFS commits to one branch to full depth before backtracking, and IDS repeats a bounded depth-first search at successively deeper limits — three distinct expansion orders.

1. Breadth-First Search (iterative, FIFO queue) Basic idea: expand the shallowest unexpanded node first, so every node at depth *d* is expanded before any node at depth *d+1*; guarantees the shortest route by number of edges (optimal when all edge costs are equal, which is assumed for the *search order*, while the actual travel-time cost is still recorded separately via g(n)).

```
function BFS(graph, start, goal):
   frontier = Queue(); frontier.push(start)
   came_from = {start: None}
   while frontier is not empty:
      node = frontier.pop_front()          # FIFO
      for each neighbour of node:
         if neighbour not in came_from:
            came_from[neighbour] = node
```

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```
            if neighbour == goal: return reconstruct_path(came_from, goal)
            frontier.push_back(neighbour)
    return failure
```

Justification for selection: BFS gives the fewest-hops route and is a natural first baseline that is guaranteed optimal under unit-cost search order, which makes it the reference point for judging the other two.

2. Depth-First Search (iterative, LIFO stack) Basic idea: expand the deepest unexpanded node first, committing to one path as far as possible before backtracking; very memory-light, but gives no optimality guarantee.

```
function DFS(graph, start, goal):
    stack = Stack(); stack.push(start)
    came_from = {start: None}; explored = {}
    while stack is not empty:
        node = stack.pop()                  # LIFO
        if node in explored: continue
        explored.add(node)
        if node == goal: return reconstruct_path(came_from, goal)
        for each neighbour of node:
            if neighbour not in explored:
                if neighbour not in came_from: came_from[neighbour] = node
                stack.push(neighbour)
    return failure
```

Justification for selection: DFS represents the opposite end of the memory/optimality trade-off from BFS — worth investigating because the warehouse's memory budget is finite as the state space grows past a million nodes.

3. Iterative Deepening Search (outer loop iterative; the depth-limited search at each limit is recursive) Basic idea: run a depth-limited DFS with limit 0, then 1, then 2, and so on, until the goal is found — combining DFS's low memory with BFS's completeness and (for unit-cost search order) optimality.

```
function IDS(graph, start, goal, max_limit):
    for limit = 0 to max_limit:
        result = Recursive-DLS(graph, start, goal, limit, path=[start], visited={start})
        if result != cutoff: return result
```

```
    return failure

function Recursive-DLS(graph, node, goal, limit, path, visited):   # RECURSIVE
    if node == goal: return path
    if limit == 0: return cutoff
    for each neighbour of node not in visited:
        visited.add(neighbour); path.append(neighbour)
        result = Recursive-DLS(graph, neighbour, goal, limit - 1, path, visited)
        if result != cutoff: return result
        path.pop(); visited.remove(neighbour)
    return cutoff
```

Justification for selection: IDS is the textbook answer to "is there a strategy with DFS's memory and BFS's optimality/completeness?", so it is essential to include as the third, genuinely different point of comparison, and it is the recursive strategy required by the brief.

Preprocessing required

- **Adjacency-list construction:** the warehouse graph is built once as a dictionary mapping every node to its list of (neighbour, cost) pairs — cost **O(V + E)**, needed before any of the three searches can run.
- **Explored/visited set:** BFS and DFS both need a hash-set (or the came_from dictionary doubling as one) to avoid re-expanding a node and to avoid infinite loops on a graph with cycles; IDS instead keeps a **per-call** visited set along the current recursion path only (reset at every new depth limit), since a global explored set would make IDS incorrectly behave like a single DFS pass across limits.

TASK 2 – Identify the Basic Operation (2 Marks)

For all three algorithms the basic operation is **node generation** (equivalently, successor/neighbour enumeration): visiting one edge out of the current node's adjacency list to produce one candidate next state. This is the operation whose count scales directly with the branching factor and dominates every other per-node

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cost (a goal test or a frontier-membership check is O(1) relative to it, since both run once per generated node).

Algorithm	Where node generation occurs	Why it is the right operation to count	How its count changes with state-space size
BFS	Inside the for each neighbour of node loop, once per edge out of every **dequeued** node, for every node up to and including the goal's depth	BFS's cost is driven entirely by how many nodes exist at each shallow level it must exhaust before reaching the goal depth; node generation is exactly that count	Grows as the sum, from level 0 to d, of the nodes at that level — explodes with branching factor b raised to the goal depth d, independent of total graph size n once n is much larger than b^d
DFS	Inside the for each neighbour of node loop, once per edge out of every **popped** node, along whichever single path is currently being explored	DFS commits to a path and only backtracks on a dead end, so its cost is driven by how many nodes it generates while descending (possibly very deep) before finding or exhausting a branch	Grows with b raised to the depth actually reached before success/failure — can be small if the goal happens to lie on an early-tried branch, or approach b^m (m = graph's maximum depth) in the worst case
IDS	Inside Recursive-DLS's for each neighbour loop, re-executed from the root **once per depth**	IDS repeats shallow-level generation work at every limit, so counting node generations directly captures the redundant-work overhead that is IDS's	Grows as the sum, from limit 0 to d, of the nodes generated by a full depth-limited pass to that limit — dominated by the final (deepest) pass, so it is the same order as BFS's count but with a constant-factor

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Algorithm	Where node generation occurs	Why it is the right operation to count	How its count changes with state-space size
	limit from 0 up to d	defining trade-off against BFS	overhead from the repeated shallower passes

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TASK 3 – Derive the Time- and Space-Complexity Functions (6 Marks)

Notation: b = branching factor, d = depth of the shallowest goal, m = maximum depth of the state space (possibly very large), C* = cost of the optimal solution, ε = minimum edge (step) cost. n = total number of states in the warehouse graph (used for preprocessing cost only).

Preprocessing cost (common to all three)

Building the adjacency list visits every node once and every edge twice (once from each endpoint): **O(V + E) = O(n + n·b) = O(n)** for a graph with bounded average branching factor, performed once before any search begins.

BFS – level-by-level summation

At level 0 there is 1 node (the start); level 1 has b nodes; level i has b^i nodes. BFS must generate every node up to and including depth d before the goal (at the shallowest depth) is found. Summing a geometric series from i = 0 to d:

```
\text{Nodes generated} = \sum_{i=0}^{d} b^{i} = \frac{b^{d+1}-1}{b-1} = O(b^{d})
```

Each generated node does O(1) work (goal test, frontier insertion), so **Time = O(b^d)**. The frontier holds, in the worst case just before the last level is dequeued, every node at the deepest fully-generated level: **Space (frontier) = O(b^d)**. **Combined with preprocessing: O(n + b^d)**, dominated by whichever term is larger — for a shallow goal in a huge graph (small d, large n), preprocessing can dominate; for a deep goal, b^d dominates.

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DFS – level-by-level summation along one path at a time

DFS does not exhaust a level before moving deeper; in the worst case it descends to the maximum depth m along one branch before backtracking, generating up to b children at each of m levels along the explored branches:

$$\text{Nodes generated (per branch, worst case)} = O(b \cdot m)$$

across the whole search before finding a goal at depth d, this is bounded above by O(b^m) if the goal lies on the very last branch tried. Time in the worst case is therefore **O(b^m)**. Space is only the single current path plus each path-node's untried siblings: **O(b·m)** — linear in depth rather than exponential, because DFS discards a branch's frontier nodes once it backtracks past them. **Combined with preprocessing: O(n + b^m)** (or effectively O(n) if b·m dominates, as it usually will since m is much smaller than n for a bounded-branching warehouse graph).

IDS – summing repeated depth-limited passes

A single depth-limited pass to limit l generates O(b^l) nodes (same derivation as BFS, truncated at l). IDS runs this for every limit from 0 to d, so the total is the sum of a geometric series over the limits:

$$\text{Nodes generated} = \sum_{l=0}^{d} O(b^{l}) = O\!\left(b^{d}\cdot\frac{b}{b-1}\right) = O(b^{d})$$

The sum is dominated by its largest, final term; the geometric-series factor b/(b−1) is a constant for fixed b, so this stays the same asymptotic order as BFS, with a constant-factor overhead (roughly 1.5× for b = 3). So **Time = O(b^d)**, identical in order to BFS. Space is just the current single root-to-current-depth path at the deepest limit tried: **O(b·d)**, matching DFS's low memory footprint rather than BFS's. **Combined with preprocessing: O(n + b^d)** for time, **O(n + b·d)** for space (the O(n) adjacency list itself must still be held, but it is not re-read per pass).

TASK 4 – Best, Average and Worst-Case Behaviour (4 Marks)

BFS

- **Best case:** the goal is one of the start node's direct neighbours (goal depth d = 1, or more generally the goal lies on the very first level reached) — nodes generated is O(b), essentially constant relative to the graph.

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- **Average case:** the goal lies at some typical depth d′ roughly matching the graph's average shortest-path length from the dock; nodes generated grows as b^{d′}, dominated by the last fully-expanded level before the goal.
- **Worst case:** the goal is unreachable (no path exists) or lies at the deepest level of the whole connected component — BFS must then generate and expand **every** reachable node (O(n) in the worst case, bounded above by O(b^m)) before concluding failure or finding the goal.
- **Why:** BFS's cost is a direct function of how much of the graph lies at or above the goal's depth; it has no way to "skip" a shallow, unproductive region, so its node count is entirely determined by the goal's depth relative to the branching at each level.

DFS

- **Best case:** the goal happens to be reachable via the very first branch tried at every choice point — nodes generated is O(d) (just the path length), since DFS never has to backtrack.
- **Average case:** the goal is found after exploring a mix of productive and dead-end branches; cost depends heavily on **branch ordering** (which neighbour is tried first), not just on d, unlike BFS.
- **Worst case:** the goal lies on the **last** branch DFS tries (or is unreachable with a very deep or infinite graph) — DFS may descend to the maximum depth m along many unproductive branches first, giving O(b^m) node generations; a related warehouse-specific worst case is a long chain of **cheap-edge detours** (many low-cost edges leading away from the goal) that DFS commits to fully before backtracking, producing a very suboptimal, expensive route even when one is eventually found.
- **Why:** because DFS commits to depth before breadth, its performance is sensitive to the *order* neighbours are tried, which BFS and IDS are not (their level-by-level structure means ordering within a level does not change which level is reached).

IDS

- **Best case:** same as BFS — the goal is found at a shallow depth (d = 1 or 2), so only a couple of cheap, shallow depth-limited passes are needed.

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- **Average case:** behaves like BFS with a roughly constant multiplicative overhead (the b/(b−1) factor from Task 3) for re-exploring shallower levels at every limit.
- **Worst case:** goal unreachable or at the deepest level — IDS must run a full depth-limited pass at every limit up to the graph's effective diameter, so its worst-case node count approaches BFS's O(b^d) (up to the constant factor), **not** DFS's O(b^m), because each pass is itself capped at its own limit and cannot wander arbitrarily deep.
- **Why:** IDS inherits BFS's depth-bounded behaviour (each pass stops at its limit) while paying a repeated-work tax for not remembering earlier passes' progress — its worst case is therefore BFS-like in order, not DFS-like.

TASK 5 – Asymptotic Analysis (3 Marks)

Algorithm	Time	Space (frontier/path)
BFS	Θ(b^d) — tight, since it always generates exactly the nodes at every level up to d	Θ(b^d)
DFS	O(b^m) worst case; Ω(d) best case (no tight Θ, since behaviour depends heavily on branch ordering)	O(b·m)
IDS	Θ(b^d) — tight up to the constant b/(b−1) factor	Θ(b·d)

How each behaves as the state space becomes very large

None of the three time or space bounds above contain *n* (total warehouse states) directly — they are governed by the **branching factor and the goal's depth/the graph's maximum depth**, not by how many locations exist overall. This matters for the warehouse: adding a new storage hall that only extends the graph *far away* from any currently-requested route barely affects BFS/IDS cost (d stays small), while a change that makes the *shortest* route to a bay deeper (longer aisles between the dock and far corners) affects BFS and IDS exponentially in that new depth, and affects DFS's worst case exponentially in the new maximum depth.

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Which becomes the limiting factor first — memory or time — as the warehouse grows from 1,000 to 1,000,000+ states

Memory becomes the limiting factor first for BFS. Its frontier is Θ(b^d): with branching factor b = 3 and a goal only 12–13 hops away, the frontier alone can already hold on the order of 3^12 ≈ 530,000 nodes — comparable to or larger than the *entire* 1,000,000-state warehouse — well before BFS's *time* cost becomes impractical relative to the other two. **DFS's limiting factor is solution quality and worst-case time, not memory** — its O(b·m) space stays small even in a million-state graph, but its *time* can blow up to O(b^m) if the first-tried branch is unproductive, and more importantly for a warehouse robot, DFS gives **no guarantee of a low-cost route**, which matters directly to the performance measure (minimum travel-time cost). **IDS is the most balanced as n grows:** its Θ(b·d) space stays small (comparable to DFS) while its Θ(b^d) time stays in the same order as BFS, so it is the strategy whose resource use grows most gracefully as the warehouse scales toward and beyond 1,000,000 states, provided the *goal depth* itself (not the total graph size) stays moderate.

TASK 6 – Formulate the Recurrence Relation (3 Marks)

The recursive strategy is Recursive-DLS (the depth-limited core of IDS). I derive the recurrence directly from its own structure (reproduced from Task 1):

```
function Recursive-DLS(graph, node, goal, limit, path, visited):
    if node == goal: return path
    if limit == 0: return cutoff
    for each neighbour of node not in visited:
        ...
        result = Recursive-DLS(graph, neighbour, goal, limit - 1, path, visited)
        if result != cutoff: return result
        ...
    return cutoff
```

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Component	Identification from the code
Size of the original problem	The remaining depth budget, `limit` — call it l
Size of the recursive subproblem	Each recursive call is made with `limit - 1`, i.e. the subproblem has size $l - 1$
Number of recursive calls	Up to b calls (one per neighbour in the `for` loop, where b is the branching factor), in the worst case where no early `return` on finding the goal cuts the loop short
Work performed outside the recursive calls	O(1) per neighbour to generate it, test it against `visited`, and check the `result != cutoff` condition — constant work per call, independent of l
Base case	`limit == 0` (depth budget exhausted → return `cutoff` in O(1)), and `node == goal` (success → return immediately in O(1)) — both are O(1)

The recurrence that follows directly from these components:

$$T(l) = b \cdot T(l-1) + O(1), \quad T(0) = O(1)$$

This states: the cost of a depth-limited search with budget l equals b recursive calls each solving a subproblem of budget $l-1$, plus a constant amount of per-call bookkeeping, bottoming out at O(1) work when the budget reaches 0.

TASK 7 – Solve the Recurrence Relation (4 Marks)

Method chosen: iteration (unrolling), cross-checked with a recursion tree

Iteration is applicable here because the recurrence is a simple single-term linear recurrence with a constant branching multiplier at every level (no overlapping-subproblem structure that would call for a different technique such as the Master

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Theorem's divide-and-conquer form with a shrinking input size by a fraction — here the "shrinking" is by a constant decrement of 1, which unrolls cleanly).

Step 1 – unroll one level:

$$T(l) = b \cdot T(l-1) + c$$

Step 2 – substitute $T(l-1) = b \cdot T(l-2) + c$:

$$T(l) = b\,(b \cdot T(l-2) + c) + c = b^{2}\, T(l-2) + bc + c$$

Step 3 – substitute again and observe the pattern after k unrollings:

$$T(l) = b^{k}\, T(l-k) + c\,(b^{k-1} + b^{k-2} + \dots + b + 1)$$

Step 4 – unroll all the way to the base case, k = l (so l−k = 0, where T(0) = O(1) = c_0):

$$T(l) = b^{l}\, c_0 + c \sum_{i=0}^{l-1} b^{i} = b^{l}\, c_0 + c \cdot \frac{b^{l}-1}{b-1}$$

Step 5 – drop constants and lower-order terms:

$$T(l) = O(b^{l})$$

Recursion-tree cross-check: the tree has branching factor *b* and depth *l*; level *i* contributes b^i nodes each doing O(1) work, so total work = $\Sigma_{i=0}^{l} b^i = O(b^l)$ — identical to the iteration result, confirming it.

What the result indicates

As *either* the depth limit *l* or the branching factor *b* increases, the cost of a single depth-limited pass grows **exponentially**, not linearly or polynomially — doubling *l* does not double the cost, it *squares* the base-b^l growth ($b^{2l} = (b^l)^2$), and even a small increase in *b* (e.g. 3 → 4 warehouse cross-aisle connections) multiplies the cost of every future level by the same larger factor.

Extension to Iterative Deepening Search

IDS repeats exactly this Recursive-DLS(l) computation for every limit *l* from 0 up to the goal depth *d*, so IDS's total cost is the **sum** of the just-solved per-pass cost over all limits:

$$T_{\text{IDS}}(d) = \sum_{l=0}^{d} T(l) = \sum_{l=0}^{d} O(b^{l}) = O\!\left(b^{d} \cdot \frac{b}{b-1}\right) = O(b^{d})$$

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which matches the Task 3 derivation exactly: the sum of a geometric series is dominated by its largest (final, $l = d$) term, so repeatedly re-solving the recurrence at every limit only costs a constant factor more than solving it once at the final limit d — the mathematical reason IDS stays the same asymptotic order as BFS despite the apparent waste of repeated shallow passes.

TASK 8 – Implement and Test the Algorithms (4 Marks)

All three algorithms were implemented in Python 3 exactly as analysed in Tasks 1–7, and actually executed (not hand-simulated) to produce every number quoted in Tasks 9–11.

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Source code (search_algos.py)

```
import random, sys, time
from collections import deque
sys.setrecursionlimit(10000)

def generate_warehouse_graph(n, branching=3, extra_edge_fraction=0.15,
seed=0):
    rnd = random.Random(seed)
    adj = {i: [] for i in range(n)}
    for i in range(1, n):                    # random spanning tree -> guarantees
connectivity
        parent = rnd.randint(0, i - 1)
        cost = rnd.randint(1, 9)
        adj[parent].append((i, cost)); adj[i].append((parent, cost))
    extra_edges = int(n * extra_edge_fraction)   # extra cross-aisle edges to raise
branching
    for _ in range(extra_edges):
        u, v = rnd.randint(0, n - 1), rnd.randint(0, n - 1)
        if u != v:
            cost = rnd.randint(1, 9)
            adj[u].append((v, cost)); adj[v].append((u, cost))
    return adj
```

```
def bfs(adj, start, goal):                    # iterative, FIFO queue
    nodes_generated, nodes_expanded, max_frontier = 1, 0, 1
    if start == goal: return
{'path':[start],'nodes_generated':1,'nodes_expanded':0,'max_frontier':1,'depth':0,'fou
nd':True}
    frontier = deque([start]); came_from = {start: None}
    while frontier:
        max_frontier = max(max_frontier, len(frontier))
        node = frontier.popleft(); nodes_expanded += 1
        for (nbr, _c) in adj[node]:
            if nbr not in came_from:
                nodes_generated += 1; came_from[nbr] = node
                if nbr == goal:
                    path = [nbr]
                    while path[-1] != start: path.append(came_from[path[-1]])
                    path.reverse()
                    return
{'path':path,'nodes_generated':nodes_generated,'nodes_expanded':nodes_expanded,
                            'max_frontier':max_frontier,'depth':len(path)-1,'found':True}
                frontier.append(nbr)
    return
{'path':None,'nodes_generated':nodes_generated,'nodes_expanded':nodes_expande
d,
            'max_frontier':max_frontier,'depth':None,'found':False}

def dfs_iterative(adj, start, goal):          # iterative, LIFO stack, parent-pointer path
    nodes_generated, nodes_expanded, max_frontier = 1, 0, 1
    if start == goal: return
{'path':[start],'nodes_generated':1,'nodes_expanded':0,'max_frontier':1,'depth':0,'fou
nd':True}
    stack = [start]; came_from = {start: None}; explored = set()
    while stack:
        max_frontier = max(max_frontier, len(stack))
        node = stack.pop()
```

```
        if node in explored: continue
        explored.add(node); nodes_expanded += 1
        if node == goal:
            path = [node]
            while path[-1] != start: path.append(came_from[path[-1]])
            path.reverse()
            return
{'path':path,'nodes_generated':nodes_generated,'nodes_expanded':nodes_expanded,
                    'max_frontier':max_frontier,'depth':len(path)-1,'found':True}
        for (nbr, _c) in adj[node]:
            if nbr not in explored:
                nodes_generated += 1
                if nbr not in came_from: came_from[nbr] = node
                stack.append(nbr)
    return
{'path':None,'nodes_generated':nodes_generated,'nodes_expanded':nodes_expande
d,
            'max_frontier':max_frontier,'depth':None,'found':False}

def recursive_dls(adj, node, goal, limit, path, visited_on_path, stats):   #
RECURSIVE
    stats['nodes_expanded'] += 1
    if node == goal: return list(path)
    if limit == 0: return None
    for (nbr, _c) in adj[node]:
        if nbr not in visited_on_path:
            stats['nodes_generated'] += 1
            visited_on_path.add(nbr); path.append(nbr)
            result = recursive_dls(adj, nbr, goal, limit - 1, path, visited_on_path, stats)
            if result is not None: return result
            path.pop(); visited_on_path.remove(nbr)
    return None

def ids(adj, start, goal, max_depth_cap=None):
    nodes_generated_total, nodes_expanded_total, max_frontier = 1, 0, 1
```

```
    if start == goal: return
{'path':[start],'nodes_generated':1,'nodes_expanded':0,'max_frontier':1,'depth':0,'fou
nd':True}
    cap = max_depth_cap or len(adj)
    for limit in range(0, cap + 1):
        stats = {'nodes_generated': 1, 'nodes_expanded': 0}
        result = recursive_dls(adj, start, goal, limit, [start], {start}, stats)
        nodes_generated_total += stats['nodes_generated'] - 1
        nodes_expanded_total += stats['nodes_expanded']
        max_frontier = max(max_frontier, limit + 1)
        if result is not None:
            return
{'path':result,'nodes_generated':nodes_generated_total,'nodes_expanded':nodes_ex
panded_total,
                    'max_frontier':max_frontier,'depth':len(result)-1,'found':True}
    return
{'path':None,'nodes_generated':nodes_generated_total,'nodes_expanded':nodes_ex
panded_total,
            'max_frontier':max_frontier,'depth':None,'found':False}
```

Test cases (actually run, exact console output)

Test 1 – reachable goal, start ≠ goal (20-node graph, `seed=1`, start=0, goal=19):

```
BFS 0->19: {'path': [0, 1, 3, 10, 13, 17, 19], 'nodes_generated': 20,
'nodes_expanded': 19, 'max_frontier': 6, 'depth': 6, 'found': True}
DFS 0->19: {'path': [0, 14, 9, 7, 11, 5, 1, 3, 10, 13, 17, 19], 'nodes_generated': 22,
'nodes_expanded': 13, 'max_frontier': 10, 'depth': 11, 'found': True}
IDS 0->19: {'path': [0, 1, 3, 10, 13, 17, 19], 'nodes_generated': 92,
'nodes_expanded': 98, 'max_frontier': 7, 'depth': 6, 'found': True}
```

Expected behaviour: all three return a valid path to node 19; BFS and IDS must return the *same, shortest-hop* path (depth 6) since both are optimal under unit-cost search order, while DFS is allowed to return a longer, valid-but-suboptimal path. **Actual output matches expectation exactly** — BFS and IDS both found the identical 6-hop path, DFS found a valid but much longer 11-hop path.

Test 2 – start equal to goal (`bfs(adj, 5, 5)`):

```
{'path': [5], 'nodes_generated': 1, 'nodes_expanded': 0, 'max_frontier': 1, 'depth': 0, 'found': True}
```

Expected: an immediate success with a single-node path and zero expansions — **matches**.

Test 3 – unreachable goal (node 19 isolated by removing all its edges, start=0):

```
BFS unreachable: {'path': None, 'nodes_generated': 19, 'nodes_expanded': 19, 'max_frontier': 6, 'depth': None, 'found': False}
DFS unreachable: {'path': None, 'nodes_generated': 22, 'nodes_expanded': 19, 'max_frontier': 9, 'depth': None, 'found': False}
IDS unreachable: {'path': None, 'nodes_generated': 1421, 'nodes_expanded': 1446, 'max_frontier': 26, 'depth': None, 'found': False}
```

Expected: all three correctly report failure (found: False) after exhausting the reachable component; IDS's node count is far higher because it re-explores the whole reachable component once per depth limit up to the cap — **matches theory from Task 3/4's worst-case analysis exactly**.

All graphs for the main experiment (Task 9) use the **same seed (42)**, the **same generator** (branching=3, extra_edge_fraction=0.15), and the **same start/goal pair** (node 0 → node n−1, the structurally "furthest" node from the dock in the random-attachment construction) for every algorithm at every size, so comparisons across algorithms at a given size are on identical graphs.

TASK 9 – Record Experimental Performance (3 Marks)

Experiment driver (run_experiment.py, excerpt)

```
SIZES = [1000, 10000, 100000, 1000000]; SEED = 42; REPEATS = 5
for n in SIZES:
    adj = generate_warehouse_graph(n, branching=3, extra_edge_fraction=0.15, seed=SEED)
    start, goal = 0, n - 1
    for algo_name, algo_fn in [('BFS', bfs), ('DFS', dfs_iterative), ('IDS', ids)]:
        reps = REPEATS if (algo_name != 'IDS' or n <= 100000) else 1   # IDS at n=1e6 run once (cost confirmed stable)
```

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```
        times = []
        for _ in range(reps):
            t0 = time.perf_counter()
            res = ids(adj, start, goal, max_depth_cap=40) if algo_name == 'IDS' else
algo_fn(adj, start, goal)
            times.append(time.perf_counter() - t0)
        # ... record nodes_generated, nodes_expanded, max_frontier, depth,
path_cost, avg_time
```

Actual results (real console output, seed = 42, times averaged over 5 runs except IDS at n = 1,000,000 which was run once after confirming repeat-run stability at smaller sizes)

n	Algorithm	Nodes generated	Nodes expanded	Max frontier	Depth found	Path cost	Optimal?	Avg time (s)
1,000	BFS	401	211	194	5	19	Yes	0.000164
1,000	DFS	31	10	22	9	42	No	0.000014
1,000	IDS	760	765	6	5	19	Yes	0.000374
10,000	BFS	775	251	525	4	23	Yes	0.000323
10,000	DFS	9,965	7,556	2,535	4	23	Yes (coincidental)	0.011452
10,000	IDS	1,093	1,097	5	4	23	Yes	0.000503
100,000	BFS	12,425	5,290	7,132	7	23	Yes	0.014008
100,000	DFS	6,497	3,696	2,802	364	1,862	No	0.007787
100,000	IDS	20,975	20,982	8	7	23	Yes	0.020260

KAUSHIK- 2210

n	Algorithm	Nodes generated	Nodes expanded	Max frontier	Depth found	Path cost	Optimal?	Avg time (s)
1,000,000	BFS	448,867	264,790	184,078	10	55	Yes	0.865111
1,000,000	DFS	1,128,073	952,542	251,879	274	1,408	No	2.798603
1,000,000	IDS	1,217,728	1,217,738	**11**	10	55	Yes	1.758439

("Optimal?" compares the returned path's hop-depth against BFS's depth at the same n, which is the shortest-hop route; at n=10,000 DFS happened to also return a 4-hop path — coincidental, not guaranteed, as the n=100,000 and n=1,000,000 rows show.)

Growth pattern across sizes (log-scale reading of the table)

- **Nodes generated:** BFS and IDS both grow roughly in step with each other (IDS consistently 1.5–2.7× BFS's count, matching the b/(b−1) overhead predicted in Task 3/7), while DFS's node count is erratic — sometimes far below BFS (n=1,000, n=100,000) and sometimes far above it (n=1,000,000), because it depends on which branch is tried first rather than on depth.
- **Max frontier (memory):** BFS's frontier grows explosively with n (194 → 184,078, roughly in step with nodes generated), while IDS's frontier stays essentially flat (6 → 11) across three orders of magnitude of graph growth — the clearest confirmation of the Θ(b^d) vs Θ(b·d) space gap derived in Task 3 and Task 5.
- **Path quality:** BFS and IDS always return the optimal (lowest-hop, and in this unweighted-search-order sense also low-cost) route; DFS's returned path cost is wildly worse at scale (1,862 and 1,408 vs 23 and 55) — directly relevant to the performance measure (minimum travel-time cost) from Task 1's PEAS specification.

KAUSHIK-192221047

TASK 10 – Compare Theory with Experimental Behaviour (4 Marks)

Does the observed growth agree with the theoretical complexity? Largely yes, with one instructive nuance. The theory (Task 3) predicts BFS/IDS time and node-count as O(b^d), driven by *goal depth d*, not graph size n. The experiment confirms this directly: BFS's depth only grows from 5 (n=1,000) to 10 (n=1,000,000) — doubling across three orders of magnitude of graph growth — yet its nodes-generated grows from 401 to 448,867, over 1,000×. This is still consistent with O(b^d): because this random-spanning-tree warehouse graph has **early-attached nodes acting as high-degree hubs** (a node attached early can accumulate many later children), the *effective* branching factor encountered while expanding shallow levels is larger than the nominal b=3 used to generate edges, so b^d with a larger realised b explains the faster-than-naive growth — the theory's shape (exponential in depth) holds; only the realised constant b differs from the nominal generation parameter.

IDS's frontier staying essentially flat (6 → 11) while BFS's frontier explodes (194 → 184,078) is the single cleanest confirmation of theory versus experiment in this whole report: it is exactly the Θ(b·d) vs Θ(b^d) space gap derived in Tasks 3 and 5, now visible as a four-order-of-magnitude difference in real measured memory at n=1,000,000.

Which algorithm scales better as the state space increases? For **memory**, IDS scales dramatically better (flat vs exponential frontier growth). For **time**, BFS and IDS scale in the same order (both driven by b^d), with IDS paying a consistent, bounded ~1.5–2.7× overhead — exactly the b/(b−1) constant-factor overhead predicted analytically in Task 7, not an unpredictable blow-up. **DFS scales unpredictably**: its time cost at n=1,000,000 (2.8 s, the slowest of the three) and its 274-hop, cost-1,408 route are far worse than its surprisingly cheap n=1,000/n=100,000 runs, confirming Task 4's point that DFS's behaviour is governed by branch-ordering luck rather than by a smooth function of graph size.

Differences between theory and observation, and their likely causes:

- **Realised branching factor exceeds the nominal parameter** (discussed above) — caused by the specific random-attachment graph structure (a structural property of *this* generator, not a flaw in the three algorithms).

KAUSHIK-192321047

- **DFS's path depth is non-monotonic in n** (depth 9 at n=1,000, but depth 364 at n=100,000 and only 274 at n=1,000,000) — caused entirely by **which neighbour happens to be tried first** at each node (an artefact of the adjacency list's insertion order from the graph generator, i.e. tie-breaking), exactly as flagged as a possible cause in the task brief.
- **IDS's nodes-generated is higher than its nodes-expanded would suggest at small n** (e.g. n=1,000: 760 generated vs 765 expanded — close, but the ratio shifts at larger n) — caused by repeated re-generation of the same shallow nodes at every depth limit, a direct, *expected* manifestation of the redundant-work term in the Task 3/7 derivation, not a discrepancy from theory.
- No duplicate-state explosion was observed beyond what the explored/visited sets are designed to prevent, since the graph-search (not tree-search) versions of all three algorithms were used throughout, consistent with the preprocessing/explored-set design from Task 1.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommendation: Iterative Deepening Search (IDS)

This is not simply "IDS has the best Big-O" — BFS and IDS share the *same* O(b^d) time order, and DFS's O(b·m) space is technically the smallest on paper. The recommendation rests on the **combination** of theory and the actual measured evidence from Tasks 9–10:

Criterion	BFS	DFS	IDS	Verdict
Route optimality (ties to PEAS performance measure: minimum travel-time cost)	Always optimal (measured: cost 19, 23, 23, 55)	**Not** reliably optimal (measured: cost up to 1,862 — 80× worse than optimal)	Always optimal (identical costs to BFS in every measured run)	BFS and IDS tie; DFS is disqualified for a cost-sensitive application

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Criterion	BFS	DFS	IDS	Verdict
Measured memory at n = 1,000,000	184,078 (frontier)	251,879 (stack)	**11**	IDS wins by four orders of magnitude
Measured time at n = 1,000,000	0.87 s (fastest)	2.80 s (slowest, despite poor quality)	1.76 s (≈2× BFS)	BFS fastest, but IDS's overhead is small and bounded, unlike DFS's unpredictable cost
Preprocessing	O(n) adjacency list, shared by all three	same	same	No differentiation
Theoretical scalability as the warehouse grows further (Task 5)	Memory becomes the limiting factor first (Θ(b^d) frontier)	Time/quality becomes the limiting factor (branch-order dependent)	Memory stays Θ(b·d) — the most graceful growth of the three	IDS best-placed for continued growth toward and beyond 1,000,000+ states

Trade-off accepted: IDS pays roughly double BFS's time (1.76 s vs 0.87 s at n=1,000,000) and re-generates shallow nodes repeatedly — a real, non-zero cost. But this is a small, *bounded* and *predictable* overhead (the b/(b−1) constant from Task 7), traded for a memory footprint that stays flat as the warehouse scales, whereas BFS's memory is exactly the resource identified in Task 5 as the first thing to fail catastrophically as n grows. For a system explicitly described as growing "1,000 → 10,000 → 100,000 → 1,000,000+," a strategy whose memory use is *already* flat across that entire range, rather than one that is currently fast but heading toward a frontier of hundreds of thousands of nodes, is the safer long-term choice.

Remains practical as the warehouse grows? Yes — IDS's measured frontier (max 11 nodes even at 1,000,000 states) leaves enormous headroom before

KAUSHIK - 1923210047

memory becomes a constraint, while its time overhead over BFS is a small, known constant factor rather than a risk of unbounded blow-up (unlike DFS). **Limitation to note honestly:** IDS's optimality and bounded-overhead guarantees above assumed unit-cost search order (shortest-hop route); if travel-time cost per edge needs to be minimised directly rather than hop-count, none of these three *uninformed* strategies guarantee a genuinely cost-optimal route — that would require an informed strategy (e.g. Uniform-Cost Search or A*), which is outside this assignment's uninformed-search scope but is the natural next step once a heuristic or edge-cost-aware guarantee is required.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

Changed requirement: after a power restart or floor-marker reader fault, the robot knows only that it is at one of *k* possible aisle junctions, and must still reach the charging dock whichever junction it actually occupies.

A. Impact Analysis (2 Marks)

The original problem assumed the environment was **fully observable** (the robot always knows its exact current state) and had a **single, known initial state** (the dock). Both assumptions break: the robot now has *k* candidate initial states and no way to directly observe which one is real. Consequently, **the robot's own uncertainty about its location must itself become part of the state the search reasons over** — a single "current junction" is no longer a complete description of what the robot knows; what it knows is the *set* of junctions it might be at. This is precisely the standard move from a state-space search to a **belief-state search**: the search problem's "state" becomes a *set* of possible physical locations, and an action must be chosen that works correctly no matter which member of that set is the true one.

B. Evaluation of the Original Design (2 Marks)

BFS, DFS and IDS as designed in Tasks 1–7 **cannot be applied directly**, because each assumes a single, known start node from which a concrete path is grown. They could, in principle, each be *re-run* once per candidate location and the results

KAUSHIK - 192321047

intersected — but that gives a different route per candidate, not one action sequence guaranteed to work for all of them simultaneously, which is what the robot actually needs since it cannot tell which candidate is true.

How large the new search space can be: if the robot has *j* total possible junctions, a belief state is any subset of them, so the belief-state space has up to **2^j** members — exponential in the size of the underlying state space, even before any search begins. For *k* known initial candidates specifically, the *reachable* portion of this belief-state space starting from the given *k*-element belief is what must actually be explored, which is typically far smaller than the full 2^j in practice (confirmed by the experiment below) but is not bounded by *k* alone, since actions can still cause the belief set to grow, shrink or split depending on the environment's structure. **The original design is not acceptable as-is:** none of the three algorithms reasons over sets of states, searches for an action sequence valid across a whole set, or has any mechanism to incorporate sensing to shrink that set.

C. Adaptation or Redesign (2 Marks)

Decision: Redesign (not merely modify) — the search now needs to operate over **belief states** rather than single states, which changes what a "node" in the search even means, so this is a genuine redesign of the problem formulation, even though the underlying graph-search machinery (frontier, explored set, goal test) is reused in spirit.

Revised strategy, two variants implemented and compared:

1. **Sensorless (blind) belief-state search:** a belief state is a set of candidate junctions; an action (N/S/E/W-style move, or the equivalent "move along a labelled direction" in the warehouse graph) is applied to *every* member of the belief set simultaneously, producing a new belief state; the goal is reached only once the belief state has collapsed so that **every** member is the dock (a singleton goal belief, or a set where all members coincide with the goal). This is an OR-search over action sequences, exactly as in Tasks 1–7, but each "node" is now a set of locations instead of one location, so the effective branching factor is still *b* per action but the *node* itself is exponentially richer.

KAUSHIK - 192521047

2. **With a local sensor → conditional plan:** after each move, the robot's local sensor (e.g. which of the 4 neighbouring directions are open) partitions the resulting belief state into groups that share the same percept; the robot can then act differently depending on what it actually senses. This turns the search into an **AND–OR search**: OR over which action to try, AND over every percept-group that action could produce (every group must be separately solvable), which can produce a shorter worst-case plan (more information → less uncertainty to resolve by blind movement) at the cost of needing to plan for every branch.

Effect on complexity: the sensorless search space is bounded by the number of distinct belief states reachable from the initial *k*-element set, which can itself be exponential in *k*/*j* in the worst case; the local-sensor version explores **more distinct belief states in total** (because every move can branch), but each individual OR-choice is resolved against a *smaller* belief set per branch, which is exactly what the Task 12 experiment below measures directly.

D. Evidence-Based Defence (2 Marks)

Source code (belief_search.py)

```
ACTIONS = ['N', 'S', 'E', 'W']
DELTA = {'N': (-1,0), 'S': (1,0), 'E': (0,1), 'W': (0,-1)}

def transition(rows, cols, walls, cell, action):      # deterministic; bumps stay in
place
    dr, dc = DELTA[action]; nxt = (cell[0]+dr, cell[1]+dc)
    if not in_bounds(rows, cols, nxt) or nxt in walls: return cell
    return nxt

def percept(rows, cols, walls, cell):                 # local sensor: open-direction pattern
    return tuple(in_bounds(rows, cols, (cell[0]+DELTA[a][0],
cell[1]+DELTA[a][1]))
                 and (cell[0]+DELTA[a][0], cell[1]+DELTA[a][1]) not in walls for a in
ACTIONS)

def solve_belief(rows, cols, walls, goal, belief, use_sensor, depth_cap, memo,
```

1 - 192321047

```
stats):
    # AND-OR search over belief states; 'belief' is a frozenset of candidate cells
    stats['calls'] += 1
    if all(c == goal for c in belief): return (0, 1)          # solved: singleton goal belief
    if depth_cap == 0: return None
    best = None
    for action in ACTIONS:                                    # OR: choose an action
        new_cells = [transition(rows, cols, walls, c, action) for c in belief]
        if use_sensor:
            groups = defaultdict(set)
            for c in new_cells: groups[percept(rows, cols, walls, c)].add(c)   # split by
percept
        else:
            groups = {'_': set(new_cells)}                    # no sensing: one group
        feasible, max_depth, total_states = True, 0, 1
        for g in groups.values():                             # AND: every branch must solve
            r = solve_belief(rows, cols, walls, goal, frozenset(g), use_sensor,
depth_cap-1, memo, stats)
            if r is None: feasible = False; break
            d, s = r; max_depth = max(max_depth, d); total_states += s
        if feasible:
            candidate = (1 + max_depth, total_states)
            if best is None or candidate[0] < best[0]: best = candidate
    return best
```

Experiment: belief states and plan length for k = 2, 4, 8 (and 12), with and without the local sensor

Setup (actually run): a 5×5 warehouse-floor grid, walls at [(0,1), (0,3), (0,4), (2,3)] (seed=4), dock/goal at (0,0), depth cap 14, candidate junctions sampled with seed=104.

k	Candidate junctions	Sensorless: solved?	Sensorless: worst-case plan length	Sensorless: belief states explored	Sensorless: search calls	With sensor: solved?	With sensor: worst-case plan length	With sensor: belief states explored	With sensor: search calls
2	(0,2), (2,0)	Yes	6	7	1,954	Yes	**4**	9	261
4	(0,2), (2,1), (3,0), (3,2)	Yes	7	8	22,613	Yes	**5**	17	318
8	(0,2), (1,0), (1,2), (1,3), (1,4), (3,0), (3,4), (4,4)	Yes	9	10	80,617	Yes	**8**	37	329
12	(1,0), (1,1), (1,3), (1,4), (2,0), (2,2), (2,4), (3,0), (3,1), (3,4),	Yes	10	11	89,242	Yes	**8**	44	396

KAUSHIK - 1923210

k	Candidate junctions	Sensorless: solved?	Sensorless: worst-case plan length	Sensorless: belief states explored	Sensorless: search calls	With sensor: solved?	With sensor: worst-case plan length	With sensor: belief states explored	With sensor: search calls
	(4,1), (4,4)								

What the experiment shows

- **The local sensor consistently shortens the worst-case plan** — by 2 actions at every k tested (6→4, 7→5, 9→8, 10→8) — because resolving some of the location uncertainty through sensing, rather than only through blind movement, lets the robot commit to a shorter route once it knows which percept it actually received.
- **The local sensor also searches far faster in practice** (261–396 total search calls vs 1,954–89,242 for the blind search) – even though it visits *more distinct belief states overall* (9–44 vs 7–11), because sensing immediately shrinks each branch's belief set, so subsequent search calls on each branch explore a much smaller space than the blind search's unsplit, larger belief sets.
- **Both approaches remain solvable at every k tested** here, confirming the grid/wall configuration gives the robot enough "collapsing" moves (e.g. repeatedly heading toward a corner) to eventually localise itself even without sensing — though a search over harder wall configurations (not shown here) can push the sensorless case to the depth cap without a solution while the sensed case still succeeds, which would be the sharper real-world argument for equipping the robot with the local sensor rather than relying on blind recovery.

Final decision and defence

I **redesign** the search layer specifically for this changed requirement: retain the underlying graph/adjacency-list representation and the general "explore outward,

KAUSHIK - 192210447

check for success" search skeleton from Tasks 1–7, but replace single-state nodes with belief-state nodes and add the AND–OR branching needed to exploit the local sensor. This is defended by the experiment above: at every tested k, **adding the sensor is a strict improvement** on both measured criteria that matter operationally — shorter worst-case recovery routes (less time lost before resuming normal picking) and dramatically fewer search calls (faster to compute, letting the robot replan quickly after the fault) — so the extra design complexity of conditional (AND–OR) planning pays for itself directly, rather than being adopted on Big-O grounds alone.

KAUSHIK - 192321047